

Intel paper

1. a. In a given RC circuit find the voltage across C and R?
b. In a given CR circuit find the voltage across R and C ?;
2. for the given expression $Y = A'B'C + A'BC + AB'C + ABC + ABC'$ realize the following
 - a. 2 input and 3 input NAND gate
 - b. 2 input and 3 input NOR gate
 - c. AND, OR, INVERTER.
 - d. INVERTER;
3. what is the importance of scan in digital system.;
4. given $A \text{ XOR } B = C$, such that prove the following
 - a. $B \text{ XOR } C = A$
 - b. $A \text{ XOR } B \text{ XOR } C = 0$;

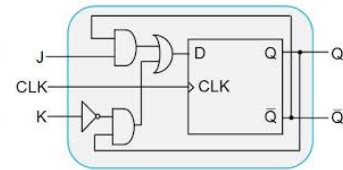
5. construct an input test pattern that can detect the result E stuck at 1 in the ckt below

$\text{NAND}(A, B) \rightarrow E$, $\text{NAND}(C, D) \rightarrow F$
 $\text{AND}(E, F) \rightarrow A$.

google:Microsoft PowerPoint - stuck_at_fault.pdf

6. in a given opamp ckt input offset is 5mv, voltage gain = 10,000, $v_{sat} = \pm 15\text{V}$ such that find the output voltage.

7. draw the p side equation of the circuit. (I am not sure about it)
8. make a JK FF using a D FF and 4-to-1 MUX.
9. use 2-to-1 MUX to implement the following expression
 $Y = A + BC' + BC(A + B)$.



10. for the following ckt what is the relation between f_{in} and f_{out} ?
the D FF use +ve edge triggered and have a initial value is 0

CLK $\rightarrow$ two DFFs with complementing (i.e one DFF have CLK and other one have

Complement of it), inputs of DFF is same and output of DFFs is given to NOR

Gate and output of NOR gate is feedback to the two DFFs.

11. design a asynchronous circuit for the following clk waveforms.

CLK $\rightarrow$ thrice the CLK period $\rightarrow$ half the period of input.

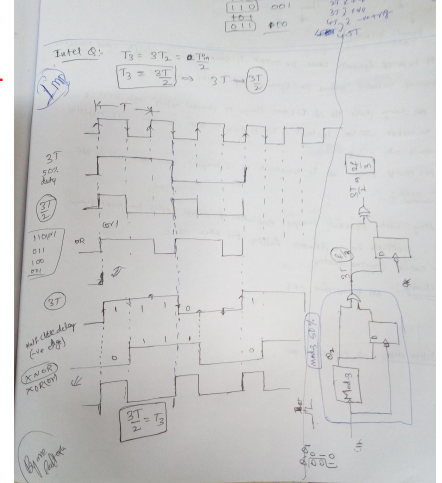
12. what is the setup time and hold time parameters of the FF, what happens if we are not consider it in designing the digital ckt.
13. given two DFF A, B ones output is the input of other and have the common clock.

F_{max} if A and B are +ve edge triggered, if A is +ve edge triggered, B is -ve edge triggered what is the F_{max} relation to previous F_{max} relation...

14. what are the FIFOs.? give some use of FIFOs in design.

15. i don't know about it but it is related to PCB layout..

technical questions are completed.
 remaining 10 questions are from optitude.
 don't worry about it.
 these can be solve easily...



ALL THE BEST

- There were 10 ques for aptitude & 16 ques for technical. Both has to be completed in 1hr.
- These are the 11 questions from them; I combined few ques and others I don't remember. If anybody remembers others pls include the same.

1. what is FIFO ? where it is used?
 2. **what is set-up and hold time?**
 3. the +ive triggered FFs are connected in series and if the maximum frequency that can operate this circuit is F_{max} . Now assume other circuit that has +ive trigger FF followed by -ive trigger FF than what would be maximum frequency in terms of the F_{max} that the circuit can work? may be $2 * F_{max}$ bcz of trigs at both edge decreases T
 4. **layout of gates were shown and u have to identify the gates (NAND & NOR gates)**
 5. make a JK FF using a mux(4:1) and a FF.
 6. the waveform of clk, i/p and o/p were shown and u have to make a sequential circuit that should satisfy the required waveform.
 7. resistor is connected in series with capacitor and the input is dc voltage. Draw the waveform across the capacitor and resistor.
 8. **two FFs, one is -ive triggered and other is +ive triggered are connected in parallel.** The 2 i/p NAND gate is has the i/p from the q_out of both the FFs and the output of the NAND gate is connected with the I/p of both FFs . Find the frequency of the output of the NAND gate w.r.t clk.
- In the **interview the questions** that they asked from me are,(which I remember). Any one if remember there question pls write it:

1. draw the circuit for inverter. How does it work.
2. if the pmos and nmos is changed in the inveretr, how does it behave.
3. design flow for ASICs and FPGA.
4. what are the difference between the ASICs and FPGA?
5. where do u use ASIC and where u use FPGA?
6. what is floorplanning?

7. what do u mean by technology file used in the synthesis or optimization for the circuit (netlist)? What is the difference in the technology files used for the ASICs and FPGAs based designing?
8. using a FF and gates. Make a memory (i.e include RD, WR etc.)
9. if the setup & hold time gets violated than what u 'll do to remove it?
10. what is clock skew? How u 'll minimize it?
11. what is clock tree? How it looks like? Concept behind that.
12. what about the Vdd and Gnd lines ? does one Vdd and Gnd pins will be sufficient for the chip. What will be the effect of using single Vdd and Gnd pins in the chip?
13. what is voltage refernce circuit? What is bandgap? How does it work?
14. what is FIFO? How does it work? Draw the circuit of FIFO of 1-bit and 4memory location deep? What would happen if memory is full and again u try to write in FIFO? What u 'll do to overcome this problem? Which one would be more easier to implement :- either dropping the packet, when the FIFO is full or pushing the data of FIFO every time. And why ?
9. given, $A \text{ exor } B = C$. Prove that: a) $A \text{ exor } B \text{ exor } C = 0$, b) $B \text{ exor } C = A$;
10. Describe Scan method? (Boundary scan etc.)
11. A CMOS circuit has given with only PDN (pull down network), you have to find the logic equation and PUN (pull up network) for the same.
12. Two questions has $Y = f(A,B,C)$. Solve for the minimum No. of gates. (I don't remember those equations)

Aptitude (10 questions)

1.next number in series

5,25,61,113,181,.....

ans:265

2>question on Avg speed

ans:100(sure)

3)question on peppers like peter,pam,pat,paul.....

ans)18(sure)

4)next number in series

1,2,4,7,.....

ans:11

5)if "FLOW" is "WOLF" then "8356".....

ans:reverse it

6)which number false in the series

64,54,42,31,20

ans:54

7)one question on odd man out

ans:c(Hexagone)

8)question regarding time

ans:10:48 a.m

9)A semi circle of radius 12,a rectangle of width 12 and length 6,
"the rectange can not be fitted in the semi circle",Is this statement
true or false

ans:False

10)

System concepts AND c,c++ programming (20+20)

1)what is ZOMBIE PROCESS?

On Unix and Unix-like computer operating systems, a zombie process or defunct process is a process that has completed execution (via the exit system call) but still has an entry in the process table: it is a process in the "Terminated state".

2)question on Argv,Argc?

argument count"; argc contains the number of arguments passed to the program. The name of the variable argv stands for "argument vector". A vector is a one-dimensional array, and argv is a one-dimensional array of strings.

Ans)4

3)Question on Least frequently used page replacement algo method,
he given some data

ans)3,4 pages

4) 8085 is a

processor 8085 is a RISC and controller 8051 is a CISC. Reduced instruction set Computer

a)cisc

b)risc

c)both

d)neither

ans:a(check it)

5)which will not spawn a process in following unix command

a)cd

b)ls

c)od

d)du

6-8 questions on bubble sort following code given 3 questions on it

```
for(i=0;i<=n-1;i++)
```

```
for(j=i+1;j<n;j++)
```

```
if(A[i]<A[j]
```

```
{
```

```
swap(a[i],a[j]);
```

```
}
```

and some data given

6)

ans)descending order

7)

ans) $n*(n-1)/2$

8)

ans)d(unchange for some values of A and N)

9)one question on segkill,segsegv,.....

10)question factorial

ans:120

11)question on critical section

a)swap(reg,reg)

b)move(reg,reg)

c)swap(reg,mem)

d)none of the above

12)question on virtual function a program given

ans) d(none of the above)

13)

```
main()
```

```
{
```

```
int ab=pqr(100*2)==10;
```

```
print("%d",ab)
```

```
}
```

```
pqr(int a)
```

```
{
```

```
return('a');
```

```
}
```

ans:0

14)thrashing is

a)more time in paging than execution

b).....

c)cpu utilization decreases with in increse of multi programming

d)both a & c

ans)d

15)char (*(x[]))() how u read it

16) another question same as above but he given sentence and we have to write expression

17)which of the following have less restrictions on database structure?

ans)1stnf

same question given two times.

18)number of nodes in balanced binary tree of height H?

ans: c(fibonacci series)

19)number of comparisons in merging two sorted lists by using merge sort?

ans)2n-1

20)address of memory depends on?

ans)number of address lines

and i didn't remember remaining questions.

Questions in the interview are like this:

*Flow of ASIC Design

*Questions in Verilog like how to generate random no.,how to get random no. between 9 and 93 etc.

*HOW the channel reduction is affected by interconnect problems andd what is the solution

*Details of setup time and hold time

*Noise separation technique

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,
>

in the second interview

>

- > 1. in which subject r u strong?
- > 2. what is difference between flip flop and a latch
- > 3. draw a circuit to divide the clock frequency by two
- > 4. draw a circuit to divide the clock frequency by three
- > 5. K map to simplify
- > 6. write a program for verilog coding for 2:8 mux
- > 7. draw the hardware of it
- > 8. can u draw the circuit which will shift 5 bits in single clock pulse
- > 9. do u know any thing about cache and dsp architecture ?
- > 10. he asked me one more question on cache i can't remember it now.

in the first interview the questions for me are

>

- > 1. what is a flip flop and what is a latch
- > 2. where do u use a flip flop and where do u use a latch
- > 3. draw a shift register
- > 4. what are setup and hold times
- > 5. do u know any thing about timing algorithms

- > 6. what are different clock routing algorithms?
- > 7. what is tree routing ?
- > and some more i can't remember right now
- >

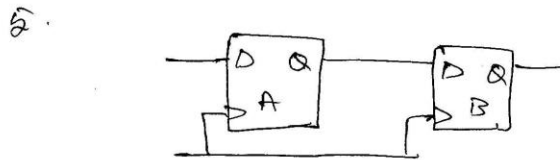
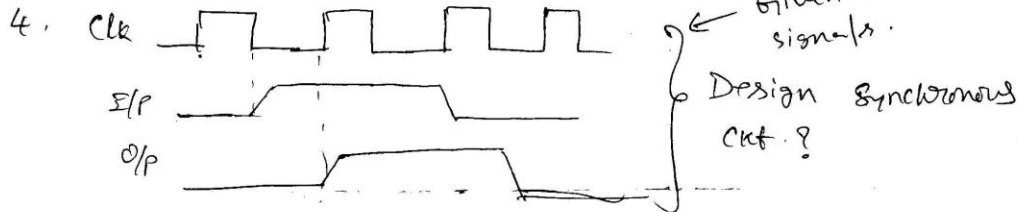
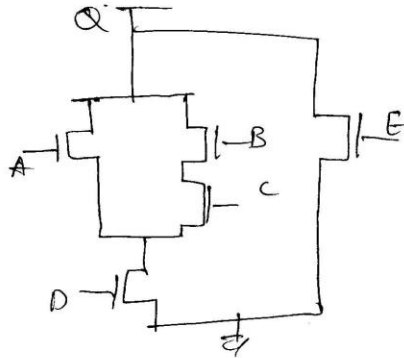
INTEL

Given
1. Layouts $\Rightarrow$ NOR & NAND

DO 2. Realization of JK F/f using one D-f/f & one 4:1 MUX

3. $Q = (A+BC)D+E$

what is Pull up network?



If Reg A & Reg B both are +ve edge trigger f/f $\rightarrow$ f_{max} .

If Reg A +ve edge & Reg B -ve edge trigger f/f $\rightarrow f_1$ then $f_1 = f [f_{max}]$?

6. What is Setup & hold time & what happens if they not met?

7. What is FIFO & applications.

8. $Y = A + BC' + C(A+B)$ realize with 2:1 MUX.

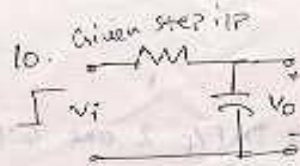


$A_0 = 10,000$

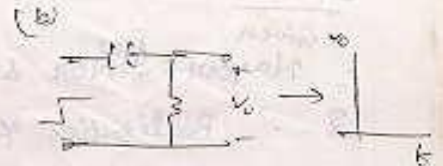
$V_{IO} = 5mV$

$V_O = ?$

+ or -15 bcz o/p cant be more than supply



Curve?

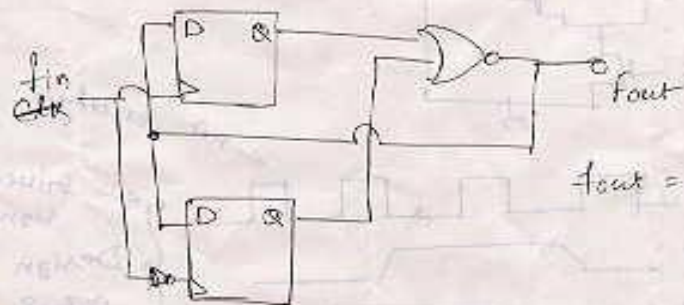


11. $f = ABC' + A'B'C + A'B'C' + \dots$

realize it with

- a) 2 i/p & 3 i/p NAND
- b) AOI & inverter
- c) 2 i/p NOR.

12.



$f_{out} = (?) f_{in}$

13.

verilog

always for reset

always @ (clk or rst)

if (rst)

d <= 1'b001

else

d <= y

end;